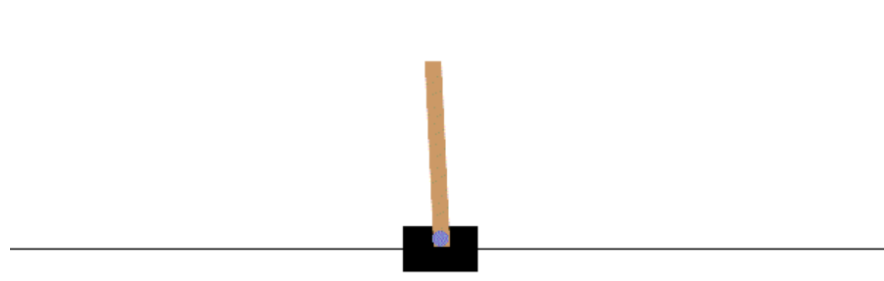


EE-568 Applied Project : RLHF

Feedback Fellows: Mathis Ayma, Diana Bejan, Yule Chen, Jiajun Hu, Mengze Tian, Arnaud Teinturier, Zhaoyu Zhu



Environments



CartPole

- State (4D): position, velocity, angle, angular velocity
- Action: Discrete $\{0, 1\}$
- Reward: +1 per step upright



Pendulum

- State (3D): $\cos \theta, \sin \theta, \dot{\theta}$
- Action: Continuous $[-2, 2]$
- Reward: $-(\theta^2 + 0.1\dot{\theta}^2 + 0.001a^2)$

Expert Training & Trajectory Generation

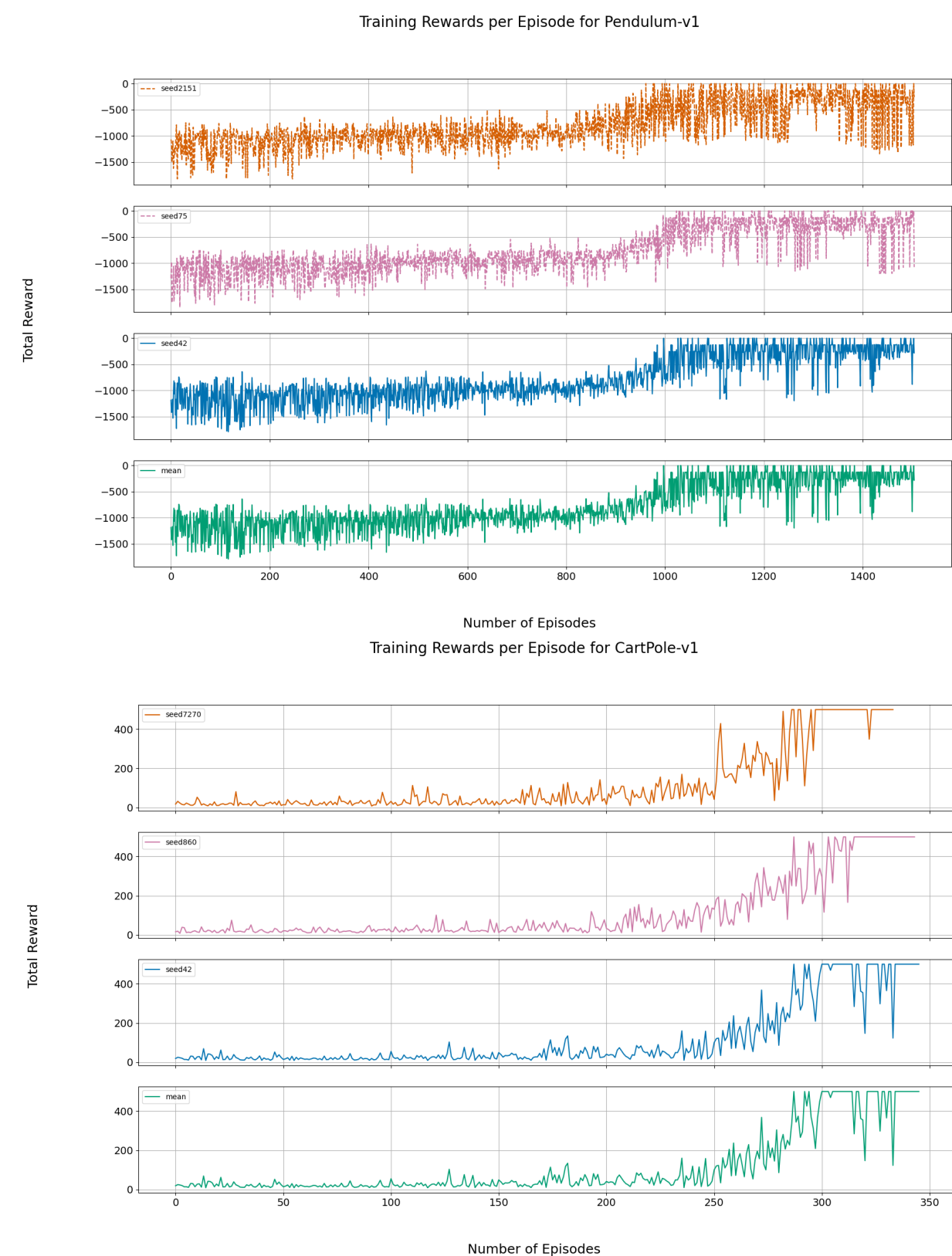


Figure 1. The consistency of the training of the expert is checked over 3 different seeds

- The expert consistently reaches the highest results, independently of the seed
- The trajectory length influenced the results of the algorithms, so trajectories that do not reach the expected trajectory length are padded with zeros

Results on Pendulum

Performance Comparison of Pendulum Environment

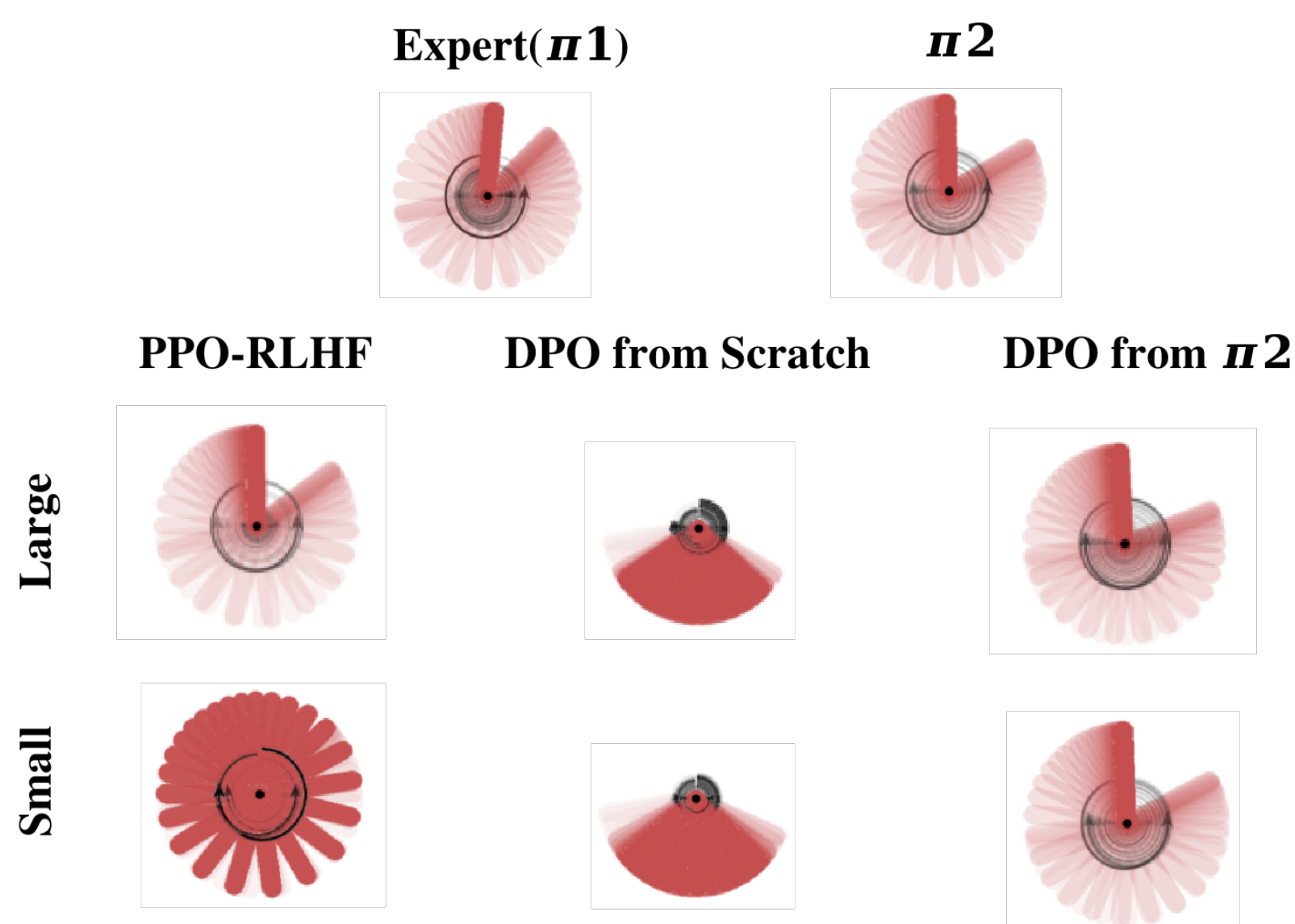
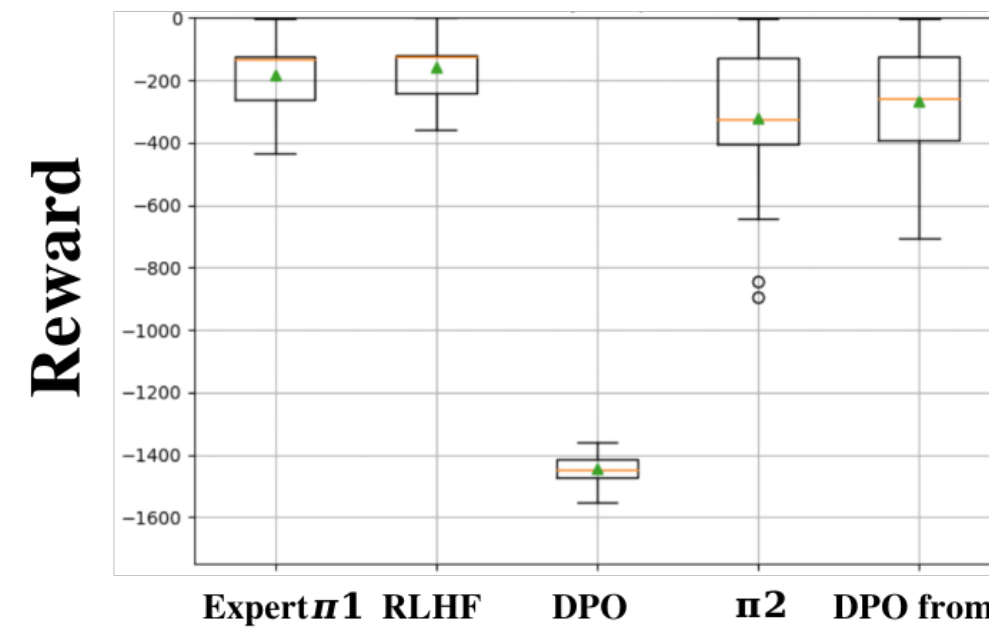


Figure 2. Performance comparison of Expert, PPO-RLHF, and DPO in the Pendulum environment. PPO-RLHF outperform DPO from π_2 when trained on large datasets. Darker colors indicate agent states with larger step magnitudes. We observe that DPO from π_2 tends to perform better when the dataset size is reduced because of reference model.

- We evaluated the effectiveness of PPO-RLHF and DPO in the Pendulum test environment, using a pre-trained expert policy as a performance reference. The comparison is conducted under large (10,000 pairs) and small (2000 pairs) dataset scales, as shown in Fig 2.
- For large dataset scale, we found PPO-RLHF outperforms the expert policy (π_1). And when DPO is fine-tuned from the slightly inferior policy (π_2), it achieves better performance than π_2 itself, which means DPO's fine-tuning works.

Large Dataset



Small Dataset

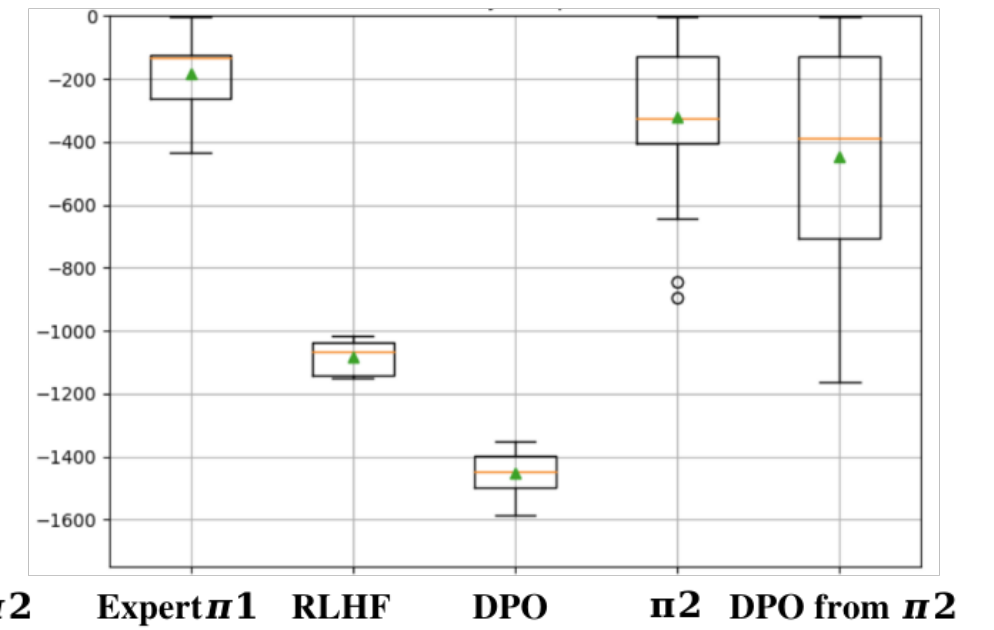


Figure 3. Benchmark of Expert, PPO-RLHF, and DPO in the Pendulum environment. PPO-RLHF exhibits greater performance compared to DPO when the dataset is large. We evaluated 20 runs using different environments to ensure statistical robustness and reproducibility of the results.

- For the small dataset, when both PPO-RLHF and DPO are trained from scratch, PPO-RLHF performs better than DPO. And DPO initialized from π_2 does not degrade significantly, since the DPO algorithm constrains the divergence between the policy and the preference model.

Results on Cartpole

Performance Comparison of CartPole Environment

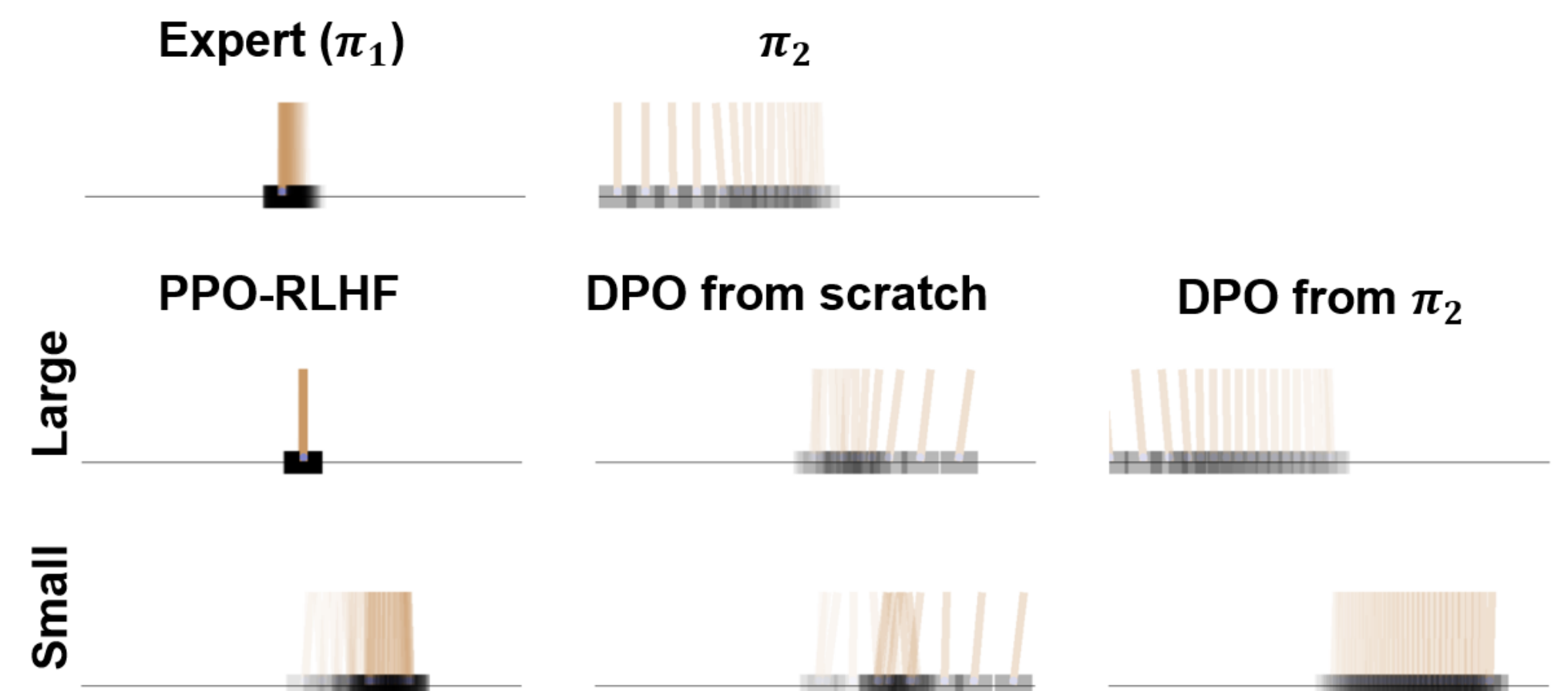
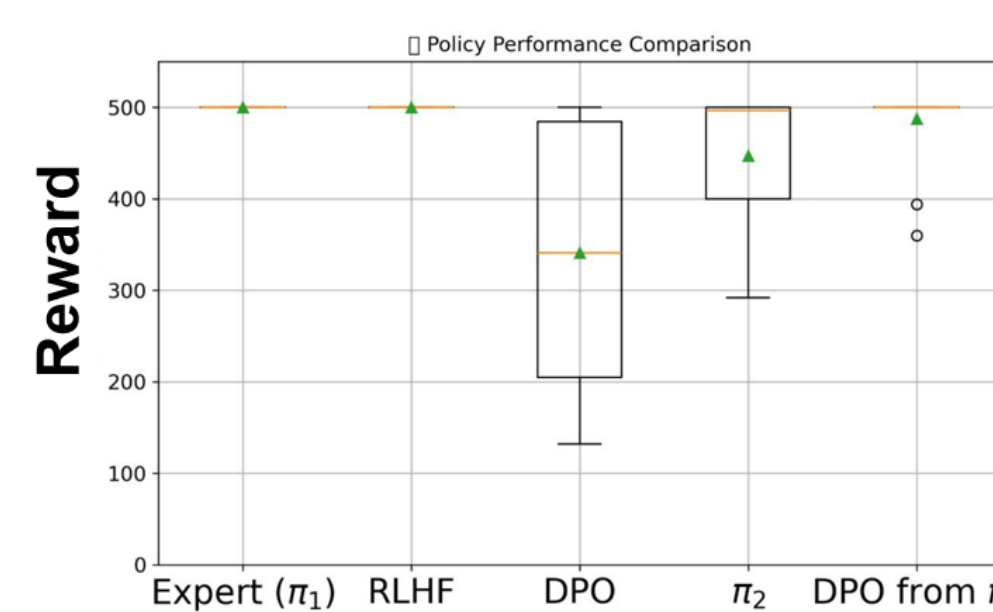


Figure 4. Performance comparison of Expert (π_1), PPO-RLHF, DPO from scratch, π_2 , and DPO from π_2 in the CartPole environment. PPO-RLHF and Expert consistently outperform DPO when trained on datasets of varying sizes. Darker colors indicate agent states with larger step magnitudes.

Large Dataset



Small Dataset

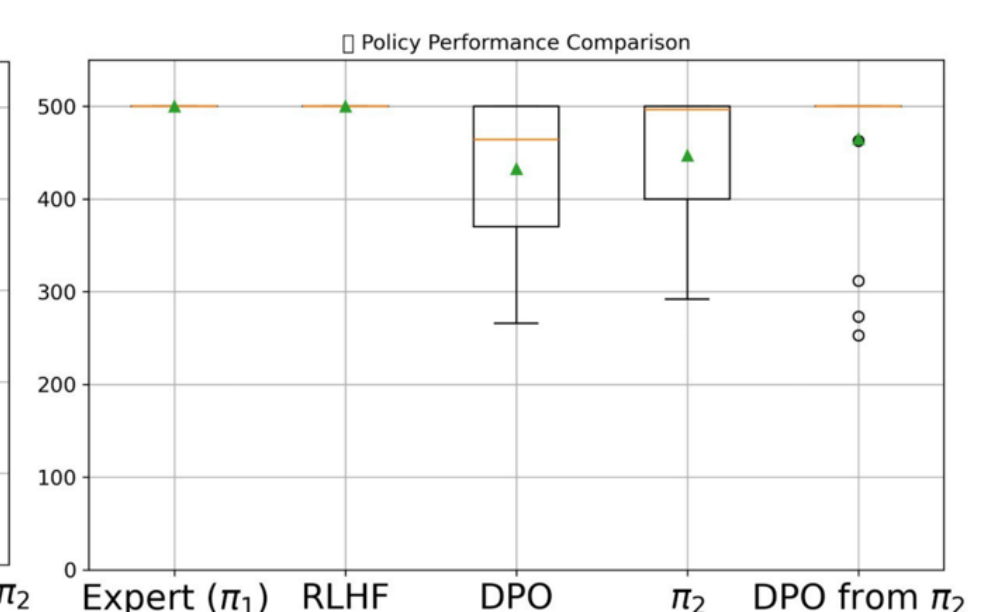


Figure 5. Benchmark of Expert (π_1), PPO-RLHF, DPO from scratch, π_2 , and DPO from π_2 in the CartPole environment. PPO-RLHF exhibits much greater stability compared to DPO, but DPO with π_2 as reference model shows better performance than π_2 or DPO from scratch.

- We evaluated the performance of PPO-RLHF and DPOs in the CartPole environment, using a pre-trained expert policy as a performance reference. As shown in Fig 5, the comparison is conducted under 2 preference dataset sizes: large (1k) and small (200). Surprisingly, DPO trained from a large preference dataset does not outperforms that trained from a small dataset.
- PPO-RLHF and DPO from π_2 outperform π_2 , achieving the highest rewards in most cases
- PPO-RLHF and two DPOs' performance in 5 are results of 3 different seeds. When seed vary, DPOs show more variance in performance, while PPO-RLHF is more stable. Specifically, DPO from π_2 is more stable than DPO from scratch, since the DPO algorithm constrains the policy's divergence from the preference model.

Appendix

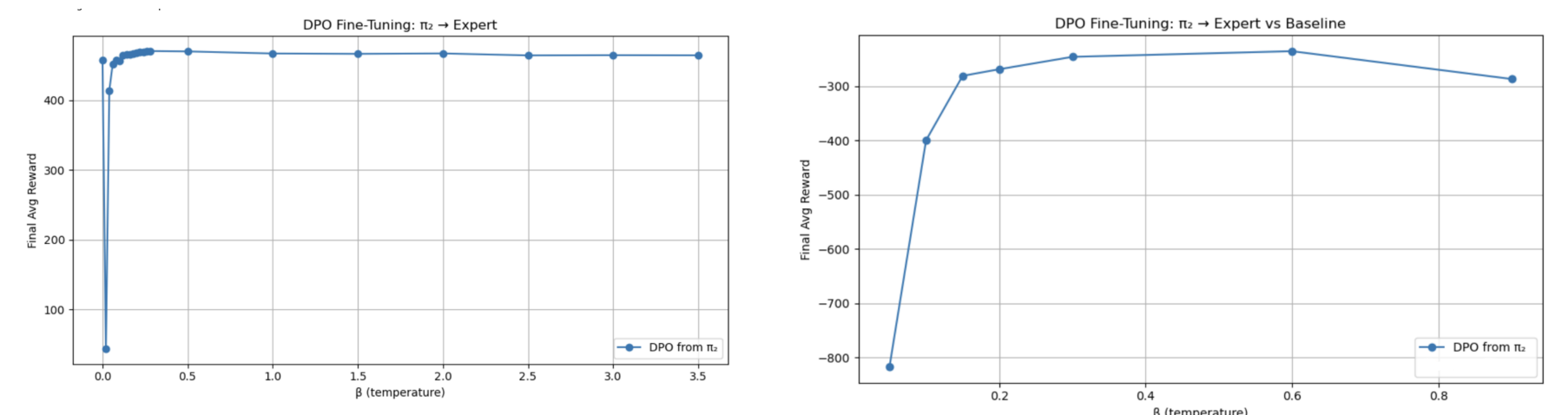


Figure 6. DPO Performance v.s. β for cartpole (left) and pendulum (right) environment on large dataset

References

- Lecture 8 of EE-568
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- Raphael Rafailov, Yiding Ge, Micah Goldblum, Tom Goldstein, Honglak Zhang, Pavel Izmailov, and Andrew Gordon Wilson. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023.